Calibrated Graph Convolutional Rollouts for Probabilistic Functional Tumor Volume Forecasting in Breast DCE-MRI

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Abstract

Longitudinal breast dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) provides repeated measurements of tumor burden during neoadjuvant treatment, but most response models reduce this sequence to a scalar endpoint or a fixed imaging feature trajectory. This work develops an endpoint-calibrated graph rollout model for forecasting patient-specific three-dimensional tumor graphs and quantifying uncertainty in future functional tumor volume (FTV). Tumors are represented as registration-consistent supervoxel graphs across T_0, T_1, T_2, and T_3. A scheduled-sampling graph forecaster predicts node displacement, feature evolution, and active-node probability. Conditional Monte Carlo simulation resamples held-out residual structure within each conditioning bucket to produce patient-level predictive distributions for future FTV and sampled tumor clouds. The retained model, `bio_ftv020_alive005`, adds aggregate FTV and alive-burden objectives to a spatial rollout baseline. On 758 held-out patients, deterministic $T_0 \to T_3$ FTV mean absolute error decreased from 12.88 mL to 5.88 mL while spatial metrics were preserved. Under the same residual Monte Carlo sampler, $T_0 \to T_3$ MC-mean FTV mean absolute error decreased from 15.26 mL to 7.61 mL, raw 90% coverage increased from 0.766 to 0.876, conformal 90% interval width decreased from 59.51 mL to 27.95 mL, Continuous Ranked Probability Score decreased from 8.35 to 5.16, and MC alive-count error decreased from 40.78 to 2.69 supervoxels. These results show that endpoint calibration is the step that turns a spatially plausible tumor rollout into a patient-level uncertainty model for future MRI tumor burden.

1 Introduction

Neoadjuvant breast cancer treatment creates a natural forecasting problem: given the tumor observed on serial MRI, estimate how much active tumor burden will remain at future treatment visits. DCE-MRI is especially relevant because it provides both morphologic and functional information about tumor response during therapy. In I-SPY studies, functional tumor volume (FTV) has been a central quantitative MRI biomarker, with prior work showing that longitudinal FTV and related DCE-MRI features carry response and recurrence information [1–3]. Public I-SPY2 and ACRIN-6698 imaging resources further make this problem testable at cohort scale [4, 5].

The modeling goal in this work is deliberately narrower and more spatially explicit than standard endpoint classification. The objective is to forecast the evolving tumor graph itself, then ask whether the forecasted graph supports calibrated patient-level FTV uncertainty. A tumor is represented as a graph of three-dimensional supervoxels whose node identities are propagated through time by deformable registration. This construction allows a graph neural forecaster to predict how each tumor region moves, changes imaging features, and remains active or inactive across visits.

A previous scheduled-sampling rollout model produced coherent three-dimensional tumor clouds, but conditional Monte Carlo diagnostics exposed an endpoint calibration failure: the deterministic rollout center was too low in FTV and active tumor mass. The uncertainty sampler then had to compensate for a biased center. This paper studies the corrected model, `bio_ftv020_alive005`, which keeps the same graph rollout backbone but adds endpoint-level FTV and alive-burden supervision. The central question is whether correcting the deterministic tumor-burden center makes the same Monte Carlo uncertainty layer sharper and better calibrated.

The contributions are:

1. an endpoint-calibrated graph rollout model for longitudinal breast DCE-MRI tumor supervoxel forecasting;
2. a conditional residual Monte Carlo protocol that produces patient-specific future FTV distributions from held-out calibration errors;
3. an empirical demonstration that FTV/alive-burden calibration reduces both point-prediction error and uncertainty width while improving raw coverage and probabilistic scoring.

2 Related Work and Novelty

2.1 Longitudinal Breast MRI and FTV

Serial breast MRI response modeling has a strong clinical foundation. I-SPY2 was designed as an adaptive neoadjuvant platform trial with serial imaging and biomarker collection [6]. Earlier I-SPY analyses established DCE-MRI and FTV as informative longitudinal measures of tumor response and longer-term risk [1, 2]. Later I-SPY2 analyses showed that combining multiple MRI features, including FTV, improves response prediction compared with using individual features alone [3]. Protocol-adherence analyses further showed that FTV performance depends on standardized acquisition and image quality, and more recent work has emphasized that longitudinal FTV estimation error itself matters when MRI is used to personalize therapy [7, 8]. These studies motivate FTV as the endpoint that the rollout distribution must be calibrated to, rather than treating the supervoxel forecast as only a geometric prediction problem.

2.2 Deep Learning for Breast MRI Response

Most published breast MRI response models operate at the patient or image level: handcrafted MRI features, radiomics, convolutional models, transformers, or multimodal predictors are used to estimate a response endpoint from one or more visits. Recent longitudinal and spatiotemporal MRI deep-learning work demonstrates the value of temporal representation learning, but still primarily maps one or more scans to a scalar response label [9, 10]. That framing is clinically valuable, but it does not directly produce a patient-specific three-dimensional tumor trajectory or a calibrated distribution over future imaging burden. The present work asks a different question: whether a model can forecast the evolving tumor graph and then quantify uncertainty in future FTV from that forecast.

2.3 Digital Twins and Image-Based Tumor Forecasting

Digital twin work is the closest conceptual neighbor. In oncology, a digital twin is generally a patient-specific model updated from individual data to simulate tumor behavior under possible interventions [11]. In breast cancer, image-based mathematical models have used longitudinal MRI to constrain reaction–diffusion and mechanically coupled tumor-growth equations for patient-specific response prediction [12–14]. More recent MRI-calibrated digital models forecast spatiotemporal triple-negative breast cancer response and evaluate alternative neoadjuvant schedules [15–17].

This paper is adjacent to that digital-twin literature but makes a different technical choice. Rather than calibrating a mechanistic PDE model of tumor cell growth and drug response, it learns a registration-consistent supervoxel graph rollout from the I-SPY2/ACRIN-derived cohort. Rather than optimizing treatment schedules, it focuses on imaging-burden forecasting across the observed MRI visits. Rather than using only a deterministic forecast, it builds empirical patient-level FTV distributions from held-out residual structure and evaluates their calibration. The result is not a complete treatment-selection digital twin; it is a data-driven imaging forecast module that could later serve as one component of a digital-twin workflow.

2.4 Graph Rollouts and Probabilistic Calibration

Graph neural networks provide a natural inductive bias for relational and dynamical systems [18, 19], and scheduled sampling addresses exposure bias in autoregressive sequence prediction [20]. This work combines those ideas with registration-consistent tumor supervoxels and endpoint-specific calibration. The probabilistic component is distinct from typical classification uncertainty and should be interpreted as empirical residual Monte Carlo, not as Bayesian posterior sampling over graph-network weights. After training, model weights are fixed; uncertainty samples are generated by resampling held-out residual structure around the deterministic rollout center. This follows bootstrap prediction-interval logic [21, 22] and related neural-network regression interval work that calibrates deterministic or probabilistic predictors using validation residuals or conformal prediction [23, 24]. The resulting FTV distributions are evaluated with coverage, conformal correction [25, 26], and Continuous Ranked Probability Score (CRPS), a proper scoring rule for full predictive distributions [27].

The contribution is the combined system: a registration-consistent graph rollout calibrated to endpoint-level FTV and alive burden, with patient-level conditional Monte Carlo uncertainty for future MRI tumor burden.

3 Methods

3.1 Cohort and Graph Representation

The analysis uses 758 patients with four longitudinal DCE-MRI visits and usable registration across T_0, T_1, T_2, and T_3. These patients come from the four-visit graph-bearing I-SPY2/ACRIN-derived cohort and are evaluated under the same five-fold held-out-patient protocol used for the scheduled-sampling baseline. Each patient is predicted by the checkpoint from the fold in which that patient was held out.

Each baseline tumor is partitioned into compact three-dimensional SLIC supervoxels [28]. Deformable registration transports the baseline supervoxel partition to later visits using symmetric diffeomorphic transforms [29]. As a result, node i denotes the same baseline tumor region at all visits, while node features are measured from the native follow-up image support.

For patient p and visit t, the graph is $G_p^{(t)} = (V_p^{(t)}, E_p^{(t)})$. Each node has six imaging-burden features: $\log(1 +$ voxel count$)$, volume in mL, percent enhancement mean and standard deviation, and signal enhancement ratio mean and standard deviation. Centroid positions are represented in millimeters after subtracting the visit-level tumor centroid. The alive target records whether a transported baseline supervoxel still has tumor support at that visit. Spatial edges use $k = 8$ nearest-neighbor connectivity within the tumor cloud. The forecaster predicts next-visit node displacement, feature change, and alive logit. Throughout the equations below, p indexes patients, t indexes visits, and i indexes supervoxels. $V_p^{(t)}$ denotes the transported supervoxel nodes for patient p at visit t, $E_p^{(t)}$ denotes the corresponding spatial edges, hats denote model predictions, and $\mathrm{FTV}_{p,t}$ denotes the observed functional tumor volume for patient p at visit t.

3.2 Endpoint-Calibrated Rollout Training

The retained model is based on the scheduled-sampling consistent graph forecaster. The architecture embeds the six node features into a 64-dimensional hidden state, applies two residual local spatial graph convolution layers, and concatenates an 8-dimensional visit-gap embedding before three per-node heads: 3D displacement, six-dimensional feature change, and alive logit. The model is initialized from the previously selected scheduled-sampling checkpoint, trained with AdamW at learning rate 10^{-4}, freezes the backbone for the first eight epochs, and trains for up to 120 epochs with early stopping.

The original one-step loss penalized node displacement, feature change, and cloud-level rollout geometry. The endpoint-calibrated model keeps those terms and adds aggregate alive and FTV losses:

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\mathrm{pos}} + 0.5\mathcal{L}_{\mathrm{feat}} + \mathcal{L}_{\mathrm{cloud}} \\ &\quad + 0.05\mathcal{L}_{\mathrm{alive\,BCE}} + 0.05\mathcal{L}_{\mathrm{alive\,mass}} + 0.20\mathcal{L}_{\mathrm{FTV}}.\end{aligned} \tag{1}$$

Here, $\mathcal{L}_{\mathrm{pos}}$ penalizes node displacement error, $\mathcal{L}_{\mathrm{feat}}$ penalizes imaging-feature error, $\mathcal{L}_{\mathrm{cloud}}$ penalizes cloud-level spatial rollout error, $\mathcal{L}_{\mathrm{alive\,BCE}}$ is the per-node alive binary cross-entropy, $\mathcal{L}_{\mathrm{alive\,mass}}$ penalizes aggregate active-node count error, and $\mathcal{L}_{\mathrm{FTV}}$ penalizes endpoint FTV error. The numeric coefficients are the retained loss weights from the biology-calibration grid. Scheduled sampling is enabled with maximum self-feeding probability 0.5 after a 10-epoch warmup and 60-epoch annealing schedule. The biology losses are annealed over 20 epochs. Checkpoint selection is still geometry-first, but adds small validation penalties for T_3 FTV error and alive-count error.

Predicted FTV is computed from the predicted node volumes and alive probabilities:

$$\widehat{\mathrm{FTV}}_{p,t} = \sum_{i \in V_p^{(t)}} \widehat{v}_{i,t}\, \sigma(\widehat{a}_{i,t}), \tag{2}$$

where $\widehat{\mathrm{FTV}}_{p,t}$ is the model-predicted FTV, $\widehat{v}_{i,t}$ is the predicted volume contribution for supervoxel i in mL, $\widehat{a}_{i,t}$ is the predicted alive logit, and $\sigma(\cdot)$ is the logistic sigmoid that converts the alive logit into an alive probability. The product $\widehat{v}_{i,t}\sigma(\widehat{a}_{i,t})$ is therefore the expected active-volume contribution of node i. FTV error is penalized in log space,

$$\mathcal{L}_{\mathrm{FTV}} = \left|\log(1 + \widehat{\mathrm{FTV}}_{p,t}) - \log(1 + \mathrm{FTV}_{p,t})\right|, \tag{3}$$

where $\mathrm{FTV}_{p,t}$ is the observed FTV. The $\log(1+x)$ transform keeps the loss well-defined at zero and makes small and large tumors contribute meaningfully. The alive-burden term penalizes aggregate active-node count error. The retained configuration, `bio_ftv020_alive005`, uses the strongest FTV/alive setting from the biology retraining grid.

3.3 Conditional Monte Carlo Simulation

For each model, each held-out patient is evaluated under three conditioning regimes: T_0 observed; T_0,T_1 observed; and T_0,T_1,T_2 observed. This yields six prediction buckets: $T_0{\to}T_1$, $T_0{\to}T_2$, $T_0{\to}T_3$, $T_1{\to}T_2$, $T_1{\to}T_3$, and $T_2{\to}T_3$.

The deterministic rollout is first computed with the held-out fold checkpoint. Residuals between predicted and observed outcomes are then pooled within each bucket, excluding the target patient. Each Monte Carlo draw samples residual components from this calibration pool:

- patient-level FTV residual,
- alive-count residual,
- global cloud displacement residual,
- local node displacement residuals,
- log-volume residuals,
- Gumbel top-k alive-mask stochasticity.

Model weights, fold assignment, observed conditioning visits, and the deterministic rollout center are fixed. Only the residual correction and alive mask realization vary across draws. Thus the raw Monte Carlo intervals estimate predictive uncertainty conditional on the trained rollout model and the empirical residual pool, while the conformal expansion provides an additional distribution-free calibration step under the usual exchangeability assumption.

Each model generates

$$758 \times 6 \times 256 = 1{,}164{,}288$$

Monte Carlo samples, corresponding to 758 patients, six conditioning buckets, and $N_{\mathrm{MC}} = 256$ draws per patient-bucket pair. Across the baseline and three biology-retrained models, the comparison covers 4,657,152 samples. Raw 90% intervals use the 5th and 95th percentiles of the sampled FTV distribution. For conformal correction, let j index a held-out calibration patient in the same conditioning bucket, let $[\ell_j, u_j]$ be that patient's raw 90% interval, and let y_j be that patient's observed FTV. The nonconformity score is

$$s_j = \max\{\ell_j - y_j,\ y_j - u_j,\ 0\}. \tag{4}$$

The empirical 90% conformal quantile of $\{s_j\}$ within the same bucket is denoted $q_{0.90}$. For a target patient with raw interval $[\ell, u]$, the conformal interval is $[\ell - q_{0.90}, u + q_{0.90}]$. Sharpness and calibration are summarized by interval width, empirical coverage, and Continuous Ranked Probability Score (CRPS). Spatial samples are evaluated with sliced Wasserstein distance (SWD) [30], Chamfer distance, and Dice overlap. Paired uncertainty for reported model differences is estimated by 5,000 bootstrap resamples over patient identities.

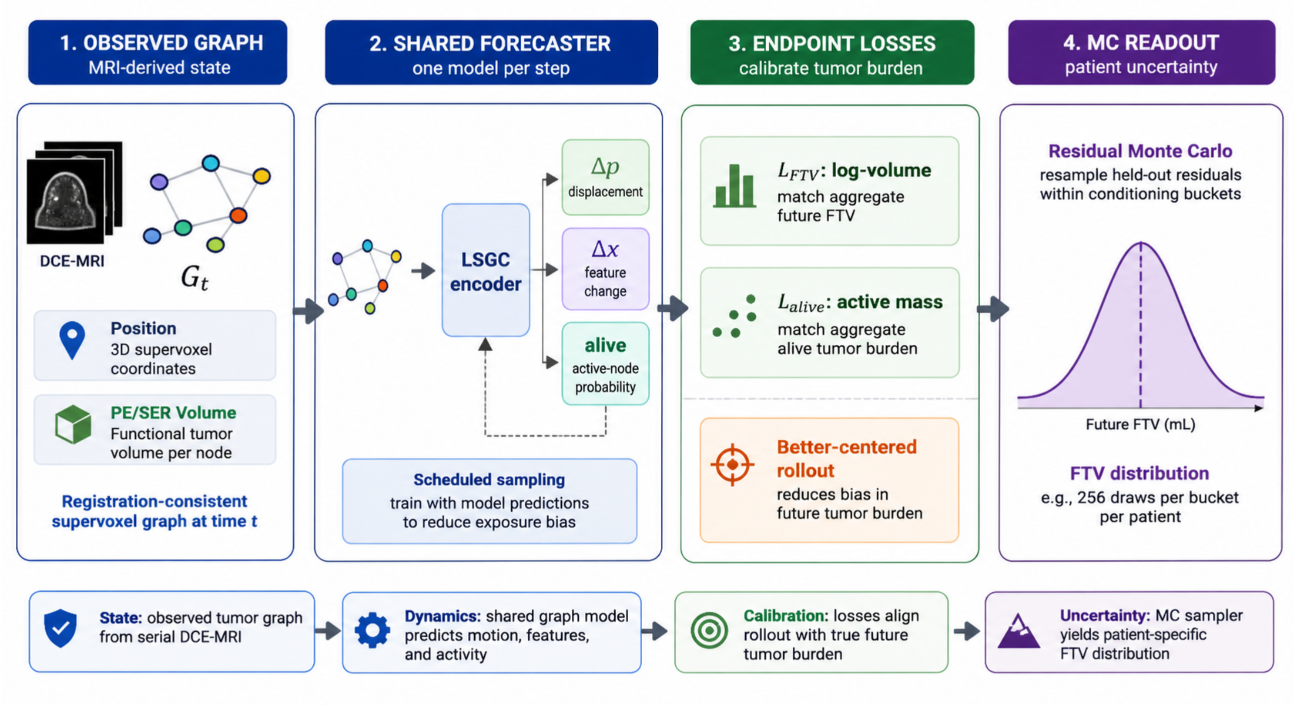

Figure 1: Forecasting and uncertainty workflow. Registration-consistent supervoxel graphs are rolled forward with a shared LSGC forecaster. Aggregate FTV and alive-burden objectives calibrate the deterministic rollout center before the unchanged residual Monte Carlo sampler generates patient-level FTV distributions.

4 Results

4.1 Endpoint Calibration Corrected the Baseline Failure Mode

The scheduled-sampling baseline had a specific endpoint failure rather than a global collapse of the graph rollout. It preserved the spatial rollout representation, but its deterministic center substantially underpredicted final tumor burden. In the held-out $T_0 \to T_3$ evaluation, observed mean FTV was 24.94 mL while the baseline deterministic mean was 12.43 mL, giving a mean bias of -12.51 mL and mean absolute error (MAE) of 12.88 mL. This is the central diagnostic result: a graph trajectory can remain geometrically plausible while still being miscentered in total active tumor volume.

Adding endpoint-level FTV and alive-burden objectives corrected that failure mode while keeping the graph rollout backbone fixed. The retained model predicted a deterministic T_3 mean of 23.21 mL, reducing bias to -1.73 mL and MAE to 5.88 mL. The paired MAE reduction was 7.00 mL (95% bootstrap CI: 6.07–7.95), a 54.3% reduction in deterministic FTV error. The retained model's $T_0 \to T_3$ deterministic bias was near zero relative to the baseline and had a bootstrap mean CI of -2.94 to -0.59 mL.

The FTV correction did not come from sacrificing tumor-cloud geometry. SWD changed from 1.291 mm to 1.294 mm, Chamfer distance from 2.837 mm to 2.843 mm, Dice from 0.1338 to 0.1331, and surface Dice from 0.4631 to 0.4622. The large change was instead in active tumor burden: alive-count absolute error decreased from 58.00 to 1.70 supervoxels. Figure 3 shows the same pattern at the patient level: endpoint calibration moves predictions toward the identity line while preserving the expected residual spread among high-volume tumors.

Table 1: $T_0 \to T_3$ deterministic endpoint results.

Metric	Baseline	Retained
Predicted FTV mean (mL)	12.43	23.21
Observed FTV mean (mL)	24.94	24.94
FTV bias (mL)	-12.51	-1.73
FTV MAE (mL)	12.88	5.88
Alive-count abs. error	58.00	1.70
SWD (mm)	1.291	1.294
Chamfer (mm)	2.837	2.843
Dice	0.1338	0.1331

4.2 The Same Monte Carlo Sampler Became Sharper and Better Calibrated

The decisive test was whether the uncertainty distribution improved when the sampler was held fixed and only the deterministic rollout center was changed. Under the same residual buckets, target-patient exclusion, alive-mask sampling, and conformal interval calculation, the $T_0 \to T_3$ MC-mean FTV MAE decreased from 15.26 mL to 7.61 mL. The paired MAE reduction was 7.65 mL (95% CI: 7.11–8.20). This shows that the baseline MC layer was not merely noisy; it was being forced to compensate for a

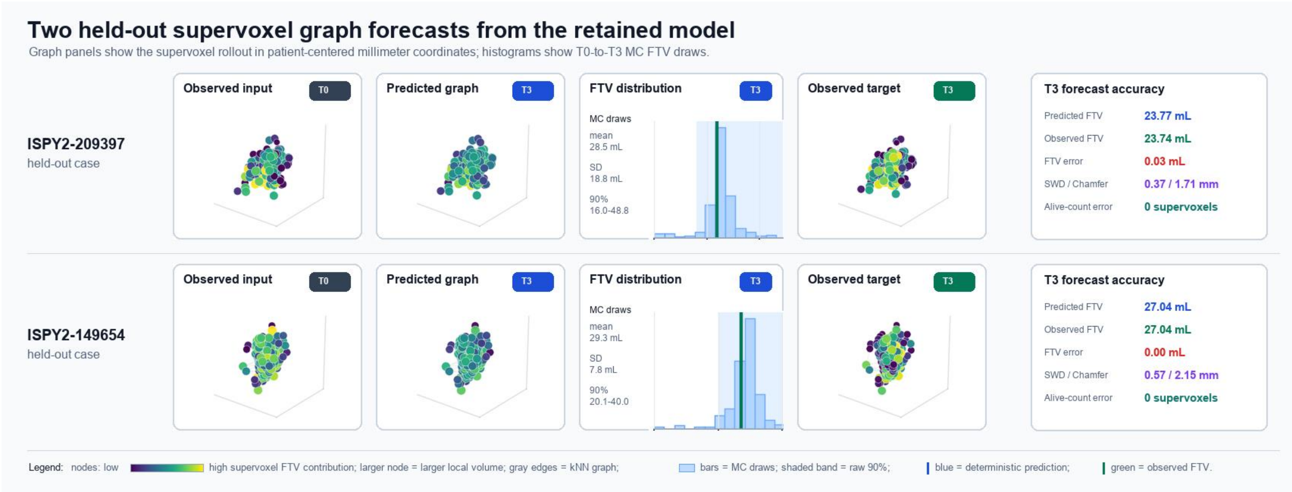

Figure 2: Representative held-out graph forecast examples generated from the retained model outputs. Each row shows the observed T_0 supervoxel graph, the retained model's predicted T_3 supervoxel graph, the $T_0 \to T_3$ Monte Carlo FTV forecast distribution, and the registered observed T_3 graph in patient-centered millimeter coordinates. Node color and size encode each supervoxel's local FTV contribution, and faint edges show the within-visit kNN graph. The retained model predicted T_3 FTV of 23.77 mL versus 23.74 mL observed for ISPY2-209397 and 27.04 mL versus 27.04 mL observed for ISPY2-149654, with 0 alive-supervoxel count error in both cases.

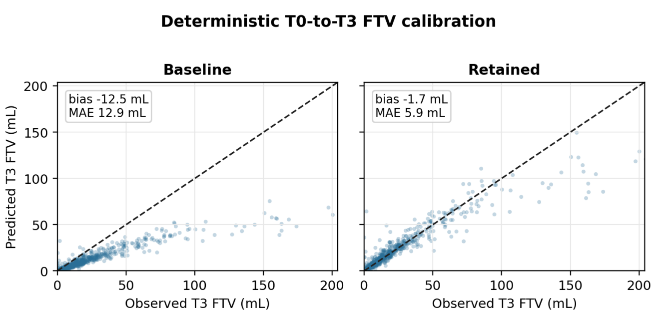

Figure 3: $T_0 \to T_3$ deterministic FTV predictions versus observed T_3 FTV. Endpoint calibration removes the large low-bias center seen in the scheduled-sampling baseline.

Table 2: $T_0 \to T_3$ conditional Monte Carlo results.

Metric	Baseline	Retained
MC-mean FTV MAE (mL)	15.26	7.61
MC-median FTV MAE (mL)	10.69	6.08
Raw 90% coverage	0.766	0.876
Raw 90% width (mL)	56.87	24.42
Conformal 90% width (mL)	59.51	27.95
CRPS	8.35	5.16
MC alive-count abs. error	40.78	2.69
SWD (mm)	1.320	1.207
Chamfer (mm)	3.656	3.485
Dice	0.047	0.051

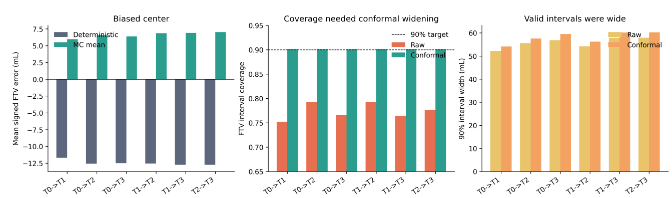

Figure 4: Monte Carlo diagnostics before and after endpoint calibration. The retained model reduces deterministic FTV bias, improves raw coverage, and substantially narrows conformal intervals.

biased center.

The interval behavior gives the strongest evidence that uncertainty calibration improved. Raw 90% coverage increased from 0.766 to 0.876 (paired increase: 0.109; 95% CI: 0.075–0.145), so the uncorrected MC distribution moved closer to nominal coverage before conformal expansion. At the same time, raw interval width decreased from 56.87 mL to 24.42 mL and conformal interval width decreased from 59.51 mL to 27.95 mL. The paired conformal-width reduction was 31.57 mL (95% CI: 30.41–32.80). The simultaneous increase in raw coverage and decrease in interval width is important: the retained model is not gaining coverage by becoming more conservative. It is producing a distribution that is both better centered and sharper.

The full probabilistic score moved in the same direction. CRPS decreased from 8.35 to 5.16, indicating that the retained distribution placed more probability mass near the observed FTV. MC alive-count error decreased from 40.78 to 2.69 supervoxels, aligning the sampled active graph burden with the volume calibration result.

4.3 Improvement Persisted Across Conditioning Times and Patients

The benefit was not limited to the baseline-to-final rollout. In the three T_3 buckets, deterministic FTV MAE decreased from 12.88, 13.17, and 13.05 mL to 5.88, 5.59, and 4.65 mL for $T_0 \to T_3$, $T_1 \to T_3$, and $T_2 \to T_3$, respectively. MC-mean FTV MAE decreased from 15.26, 14.88, and 13.80 mL to 7.61, 7.18, and 5.80 mL. The paired MC-mean MAE reductions were 7.65 mL (95% CI: 7.11–8.20), 7.69 mL (95% CI:

Table 3: $T_0 \to T_3$ Monte Carlo ablation comparison. The high-volume tail is the top decile of observed T_3 FTV ($\geq$58.79 mL; 76 patients). SV denotes supervoxels.

Model	Det. bias (mL)	Det. MAE (mL)	MC MAE (mL)	Raw cov.	Conf. width (mL)	CRPS	Alive err. (SV)	Tail MC MAE (mL)
Baseline	-12.51	12.88	15.26	0.766	59.51	8.35	40.78	42.59
FTV .020 + alive .005	-1.73	5.88	7.61	0.876	27.95	5.16	2.69	29.65
FTV only .010	-1.51	5.99	7.31	0.875	26.92	5.15	7.00	30.85
FTV .010 + alive .002	-1.50	5.99	7.53	0.883	29.14	5.19	2.70	29.79

7.17–8.23), and 8.00 mL (95% CI: 7.46–8.59), respectively. Raw coverage increased from approximately 0.76–0.78 to 0.86–0.88, while conformal widths decreased from approximately 60 mL to 24–30 mL. The paired conformal-width reductions were 31.57, 30.53, and 36.06 mL for the three final-visit buckets.

Figure 5 summarizes this result across the final-visit conditioning regimes. As more observed MRI visits are available, all methods improve, but the retained and retrained ablations remain separated from the baseline across MC-mean MAE, conformal width, and CRPS. This is important because the result is not a single $T_0 \to T_3$ artifact; the same endpoint-calibrated center improves future FTV uncertainty when the model is conditioned from baseline, early treatment, or mid-treatment imaging.

The improvement was also consistent across molecular subtypes in the $T_0 \to T_3$ bucket. MC-mean FTV MAE reductions were 7.56 mL for HR+/HER2+ tumors, 7.51 mL for HR+/HER2- tumors, 7.56 mL for HR-/HER2+ tumors, and 7.89 mL for triple-negative tumors. The smallest subgroup was HR-/HER2+ ($n = 57$), but even there the retained model reduced MC-mean FTV MAE from 18.34 to 10.78 mL. This subgroup remained the hardest case, so the result supports broad improvement but not uniform calibration across all biologic strata.

The two uncertainty visualizations in Figures 6 and 7 support complementary claims. The cohort-sorted bands show population-level behavior: across the held-out $T_0 \to T_3$ cohort, endpoint calibration shifts the MC mean toward the observed T_3 burden and compresses the predictive band, especially outside the lowest-volume cases. The patient trajectory examples then show the same mechanism at the individual level. Those panels span low, median, high, and large-improvement T_3 cases, so they are not a separate performance estimate; they are a diagnostic of how the Monte Carlo distribution changes over visits. Together, the figures show that the retained model provides a patient-specific distribution over the future MRI tumor-burden trajectory rather than only a single final-visit point estimate.

5 Discussion

This study shows why endpoint calibration is not an optional refinement for longitudinal imaging rollouts. The baseline model answered a spatial question: can a registration-consistent graph forecaster roll a tumor cloud forward through treatment? The conditional MC experiment asked a more demanding question: can that forecast serve as a calibrated center for uncertainty in future active tumor burden? The baseline answer was no. Its tumor clouds were plausible enough to support rollout evaluation, but the FTV center was biased low and alive mass was poorly calibrated. Conformal correction could recover nominal coverage, but only by expanding already wide intervals around a miscentered forecast.

The retained model changes the interpretation of the whole pipeline. With the sampler, residual buckets, held-out fold protocol, and conformal calculation unchanged, endpoint-calibrated training produced sharper and better-centered distributions. That isolates the mechanism: the main bottleneck was not the Monte Carlo machinery itself, but the deterministic rollout center used by that machinery. A residual uncertainty layer cannot turn a systematically underestimated tumor-burden trajectory into a clinically interpretable predictive distribution unless the base forecast is first calibrated to the endpoint being queried.

This matters because the target use case is not only to animate plausible tumor motion. The model should answer imaging-burden questions at the patient level: given the MRI visits observed so far, what future FTV distribution is expected, how wide is the plausible range, and does the observed endpoint fall inside a calibrated interval? The retained model answers those questions with lower error and narrower uncertainty than the original baseline. The trajectory examples are important in this context because they show the distribution as a longitudinal patient course rather than as an isolated T_3 summary; the uncertainty object is a forecasted tumor-burden path, not only a final scalar. It therefore moves the framework from spatial graph forecasting toward an imaging digital-twin component: a patient-specific, measurement-aware model of future MRI tumor burden.

Table 3 also clarifies why `bio_ftv020_alive005` is the strongest current candidate rather than simply the lowest-MAE ablation. The FTV-only model had slightly lower $T_0 \to T_3$ MC-mean MAE (7.31 mL versus 7.61 mL), but its MC alive-count error was substantially worse (7.00 versus 2.69 supervoxels). The retained model gives the better scientific balance: strong deterministic FTV accuracy, large alive-burden correction, improved raw coverage, narrower intervals, lower CRPS, better high-volume tail performance than the FTV-only ablation, and preserved spatial metrics. That balance is essential because the method is intended to remain a graph rollout model, not collapse into a scalar FTV regressor.

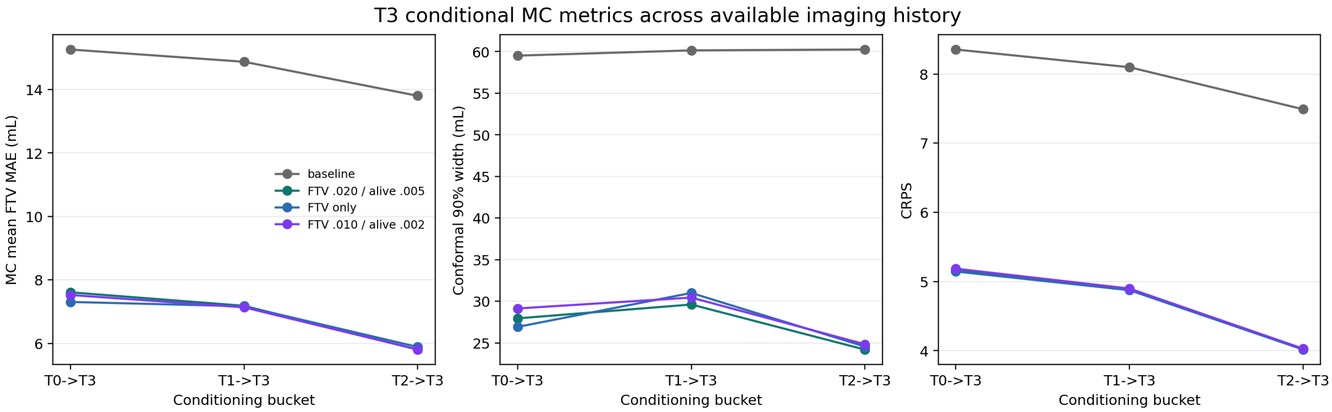

Figure 5: Final-visit conditional MC metrics across available imaging history. Endpoint-calibrated models reduce MC-mean FTV MAE, conformal interval width, and CRPS across $T_0 \to T_3$, $T_1 \to T_3$, and $T_2 \to T_3$ conditioning.

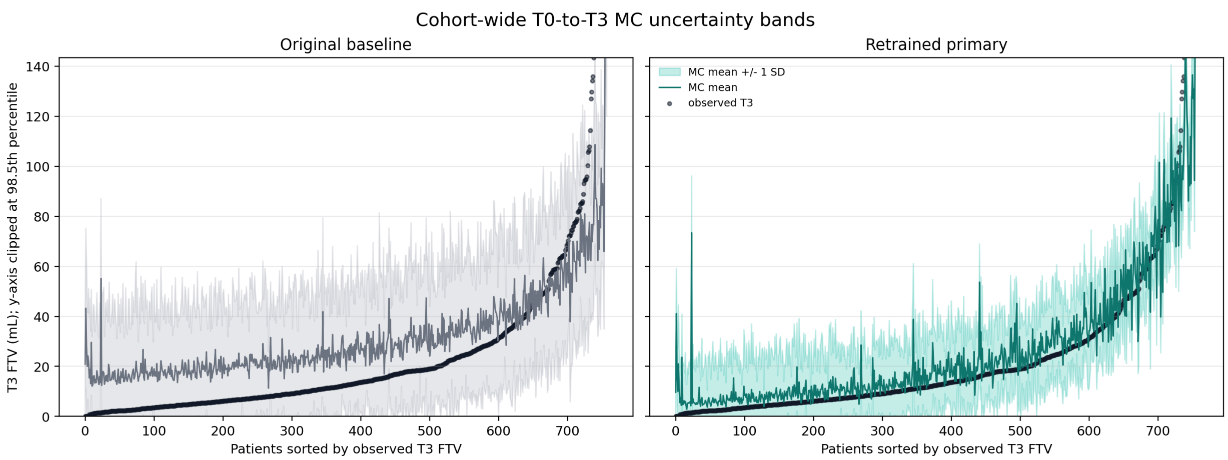

Figure 6: Cohort-wide $T_0 \to T_3$ MC uncertainty bands, with patients sorted by observed T_3 FTV. Endpoint calibration produces a narrower and better-centered patient-level distribution while improving raw coverage and CRPS.

The remaining spatial results should be interpreted carefully. Endpoint calibration did not materially degrade SWD, Chamfer distance, or Dice, which is necessary for preserving the graph-forecasting claim. However, the largest gains are clearly in tumor-burden calibration rather than in full spatial uncertainty. The current evidence therefore supports a calibrated FTV uncertainty model with preserved cloud geometry, not a complete solution to probabilistic three-dimensional tumor-shape forecasting.

5.1 Limitations

This is still an imaging-burden forecasting model, not a complete clinical decision model. It predicts future MRI FTV and sampled tumor clouds under the available graph representation; it does not model treatment counterfactuals, optimize therapy schedules, or predict endpoints that are not observed in the longitudinal MRI sequence. The residual Monte Carlo sampler is empirical and depends on the exchangeability of held-out residuals within conditioning buckets. High-volume T_3 tumors remain the hardest subgroup, suggesting that future residual sampling may need FTV-stratified or subtype-aware calibration. The current evaluation is internal to the I-SPY2/ACRIN-derived cohort; external validation on independent longitudinal breast MRI cohorts is required before clinical translation.

6 Conclusion

Endpoint calibration converted a strong spatial rollout model into a much more useful probabilistic tumor-burden forecaster. The retained model, `bio_ftv020_alive005`, reduced $T_0 \to T_3$ deterministic FTV MAE by 54%, halved MC-mean FTV MAE, improved raw 90% coverage, cut conformal interval width by more than half, and reduced MC alive-count error by more than 90%. The central result is methodological and practical: for longitudinal MRI graph rollouts, uncertainty estimates become meaningful only after the deterministic forecast is calibrated to patient-level active tumor burden. Once that calibration is in place, the model can support the imaging question that motivated the MC experiment: what distribution of future MRI tumor burden should be expected for this patient,

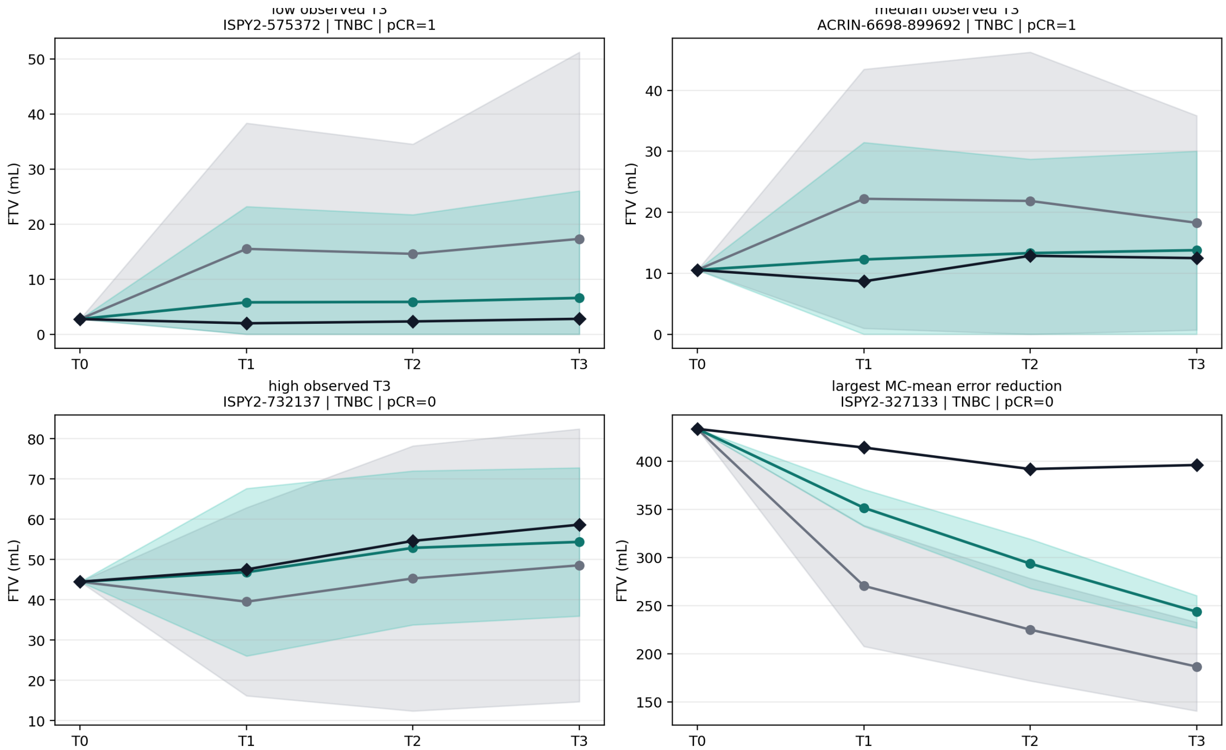

Figure 7: Representative $T_0 \rightarrow T_3$ Monte Carlo trajectory uncertainty examples. Gray shows the baseline MC mean and standard-deviation band, teal shows the retained-model MC mean and standard-deviation band, and black diamonds show observed FTV.

given the visits observed so far?

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